# Autoregressive Modelling for Streaming Time Series Data

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Accurate prediction of passenger load is a critical part of Intelligent Public Transport Systems (IPTS), and can be useful for both public transport agencies and passengers. Autoregressive models were trialled on a public transport dataset from Salvador, Bahia, Brazil, in a streaming setting with delayed labelling. The regression target was the average 5-minute load over the next 30 minutes, with metrics calculated across ten different bus stops across Salvador. Principles from Autoregressive Integrated Moving Average (ARIMA) time series analysis models were adapted to the streaming setting.

Key experiments performed included building individual autoregressive models for each station, building combined autoregressive models for all bus stops, predicting using differenced load values, and including exogenous variables for prediction. Exogenous variables that were used include weather data, Google Search trend data, and levels of flu in the population.

The best model was an integrated autoregressive model that included an embedding of the time of day and the station. The mean absolute error (MAE) of this model was 6.24 passengers. This model was applied to a station recommendation system that is able to predict the load at a given station, and divert passengers who wish to avoid crowded stations to a nearby non-crowded station.

CCS Concepts: • **Computing methodologies** → **Time series analysis**; • **Mathematics of computing** → *Time series analysis*; • **Information systems** → Decision support systems.

Additional Key Words and Phrases: Time series forecasting, streaming data, multivariate prediction, ARIMA, public transport, passenger volume, online regression, smart cities

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## 1 Introduction

Increasing urbanisation around the world has resulted in a rising demand for public transport services, particularly in developing countries. However, financial and infrastructure constraints have made it difficult for governments to match this growing demand [Motta et al. 2013]. The public transport crowding that has resulted from this causes increased stress, greater unreliability of public transport services, and reduced productivity for passengers [Tirachini et al. 2013].

Intelligent Public Transport Systems (IPTS) is a nascent field that attempts to address issues with public transport using machine learning (ML) [Zear et al. 2016]. Accurate prediction of public transport volumes is one way that ML can be applied to improve public transport and reduce crowding. If passenger volumes are known ahead of time, it may be possible for public transport agencies to optimise the availability of buses and trains in order to match demand and thus reduce crowding. Passengers can also directly benefit from prior knowledge of the demand at a given station or bus route, as they can plan their travel to avoid crowded stations [di Torrepadula et al. 2024].

The data stream mining paradigm is designed for large quantities of data which arrive continuously, in a setting where memory and computational power might be constrained [Bifet et al. 2023]. Public transport data fits this paradigm, thus the field of data stream mining provides us with a set of ML algorithms that are appropriate for deployment in this context.

The setting of this study is Salvador, the fifth largest city in Brazil. Compared to other metropolitan areas in Brazil, there is a higher reliance on public transport in Salvador due to a broadly lower socioeconomic status. The majority of public transport in Salvador is by bus, although some passengers take the Salvador Metro [Harrison and Yang 2016]. The Salvador Urban Network Transportation (SUNT) dataset contains high granularity information about passenger volumes at bus stops and stations across Salvador. The datasets include boarding, alighting, and bus occupancy spreadsheets, which contain accumulated data at 5-minute and 30-minute intervals. The time range spans from 05:00:00 on 1 March 2024, to 00:55:00 on 31 May 2024. Data are provided for 10 bus stops across Salvador, listed in Table 1 [dos Santos Ferreira et al. 2025].

Table 1. Bus stop locations

| ID | Location |
|---|---|
| 44783654 | Federal University of Bahia (UFBA) |
| 43768720 | Lapa Station |
| 230565994 | Itapuã Lighthouse Beach |
| 125960550 | Arena Fonte Nova |
| 45833547 | Manoel Barradas Stadium |
| 44784438 | Ferry Boat terminal |
| 47568123 | Near a major shopping mall |
| 44072192 | Castro Alves Theater |
| 258781031 | Salvador Bus Station |
| 44783914 | Lacerda Elevator tourist area |

The aim of this study is to investigate data stream ML models by which the load of passengers on buses in Salvador can be estimated, in order to generate as accurate a prediction as possible. There are several practical uses for such a model, including:

- Public transport agencies could estimate the number of buses that need to be deployed to particular stations at a given time to ease overcrowding.

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- Individuals could use models to estimate which stations are crowded at a given time, and get alternatives for stations that are not crowded.
- Explainable AI techniques could be used to gain insights into factors influencing public transport.

## 2 Related Work

Much of the previous work conducted on data stream time series models involves adapting ARIMA (Autoregressive Integrated Moving Average) models from the batch setting to the streaming setting [Anava et al. 2013; Liu et al. 2016]. This generally involves increasing the order of the autoregressive portion and removing the moving average portion, which is often unstable in a streaming setting [Kozitsin et al. 2021].

The vast majority of previous work done on estimating passenger load at a given station has been in the batch setting. Techniques that have been employed have included ARIMA and its extensions such as SARIMA, support vector machines, neural networks, Bayesian networks, autoencoders, recurrent neural networks, graph neural networks, and transformers. Besides autoregressive features, several exogenous features have been employed to predict public transport load including the weather, holidays, events, population in a given area, air quality, and web search volumes [di Torrepadula et al. 2024].

Little literature has been published looking at estimating passenger load in the online setting. [Zhong et al. 2023] employed principal component analysis (PCA) in order to produce a representation that could be classified using a linear regressor.

## 3 Proposal

This study will aim to trial several different data stream time series approaches towards predicting the load of passengers at each station. The aim will be to produce a model with the lowest mean absolute error (MAE) for passenger volume prediction.

The key differences between our approaches and standard time series prediction are:

- Our model should work in a streaming setting with delayed labelling.
- Our model should be able to handle and take advantage of multivariate time series (i.e. multiple stations).
- Our model should be able to predict an averaged target.

Our approaches will primarily draw from the ARIMA paradigm, and its extended formulations. We aim to explore autoregression (including non-linear autoregression), seasonality, integrated models, and exogenous regressors.

As a proof-of-concept, we will apply our best model to a prototype station recommendation system which predicts crowding at stations and provides real-time suggestions for alternative stations. In this way, passengers can avoid overcrowded stations, thereby reducing congestion and improving travel efficiency.

## 4 Formulation and Data Processing

The primary dataset used for this study was the SUNT passenger load dataset which estimates the bus occupancy at each of the 10 stops defined in Table 1, at 5-minute intervals. The target to be predicted at each time step was the average of 5-minute loads at that stop for the next 30 minutes (i.e. the average of the next six load values). This averaged target will give a gauge of load at each station over the next 30 minutes, which will give passengers or authorities enough time to respond to a prediction (e.g. by choosing a different station or sending out more buses).

Concretely, given load $x_{s,t}$ being the observed load at stop $s$ at time step $t$, the target $y_{s,t}$ to be predicted at each time step is defined as:

$$y_{s,t} = \frac{1}{6} \sum_{i=1}^{6} x_{s,t+i}$$

Autoregression will form the basis of all experiments. To predict $y_{s,t}$, $k$ previous values of $x_{s,t}$ will be taken from previous time steps, up to and including the current load, where $k$ is the order of the autoregressive model:

$$\{x_{s,t-k+1}, x_{s,t-k+2}, \ldots, x_{s,t}\}$$

We aim to learn a model $f$ which maps the autoregressive features to the target $y_{s,t}$. For our experiments, we used $k = 5$.

### 4.1 Exogenous Datasets

Contextual information relevant to Salvador and Bahia was sourced and added to the dataset. This included:

- Whether there was a public holiday in Brazil, or a significant cultural event happening in Salvador.
- Daily count of dengue cases and flu-like symptoms from OpenDataSUS [Ministério da Saúde nd]
- Opening stock value of the Bovespa Index, Brazil's main stock index, from the Yahoo Finance API. Imputed with the most recent opening stock price if none is available for that hour.
- Hourly weather data from Salvador, including temperature, wind speed, precipitation, and cloud cover. From the Open-Meteo API.
- Hourly pollution data from Salvador, including air quality index. From the OpenMeteo API.
- Daily Google Searches for "ônibus" (bus), "metrô" (subway), "festival", "trem" (train), "greve" (strike/protest). From Google Trends.

Contextual information was appropriately lagged such that the data aligned with each time step represents the most recent data available at that time step (i.e. daily information was lagged by one day, and hourly information was lagged by one hour). This simulates the real-time availability of this information.

## 5 Experiments

All experiments were performed in Python using CapyMOA 0.8.2 [Gomes et al. 2025]. All results and diagrams are reproducible using the provided notebook.

The two primary evaluation metrics we aimed to optimise were mean absolute error (MAE) and root mean square error (RMSE). These were calculated across all bus stops across all time steps:

$$\text{MAE} = \frac{1}{ST} \sum_{s=1}^{S} \sum_{t=1}^{T} \left| \hat{y}_{s,t} - y_{s,t} \right|$$

$$\text{RMSE} = \sqrt{\frac{1}{ST}\sum_{s=1}^{S}\sum_{t=1}^{T}(\hat{y}_{s,t} - y_{s,t})^2}$$

A custom prequential evaluation function was used to simulate the six time step delay in labelling. At each time step, an instance was tested, and then the model was trained on the instance from that stop from six time steps previously. This was used to calculate overall metrics. Windowed evaluation was also performed using a sliding window of 2400 instances.

K-Nearest Neighbours, Fast Incremental Model Tree with Drift Detection (FIMT-DD) [Ikonomovska et al. 2011], and Adaptive Random Forests (ARF) were the key base regressors used for each approach. Each of these regressors was evaluated with every modelling approach.

### 5.1 Baseline Naive Estimator

As a baseline estimator, the average of the last five load values was used to estimate $y_{s,t}$:

$$\hat{y}_{s,t} = \frac{1}{5}\sum_{i=0}^{4} x_{s,t-i}$$

### 5.2 Station + Hour Time-Aware Baseline

A simple model was trained to predict the average 30-minute load using only the current hour $\mathbf{h}_t \in \{0,1\}^{24}$ and station ID $\mathbf{s} \in \{0,1\}^S$, with no autoregressive or external features. The goal was to test whether temporal and spatial patterns alone carry predictive value.

One-hot encodings of the hour and station were used as inputs. The model learns a function $f$ such that:

$$\hat{y}_{s,t} = f(\mathbf{h}_t, \mathbf{s})$$

### 5.3 Per-Stop Autoregressive Models

Ten individual models $f_s$ were trained for each of the ten stops. Two variants of this approach were tested:

*Variant 1:*

In this variant, the lag values from all ten stops were included as inputs for each of the 10 models.

$$\mathbf{x}_t = \left[x_{1,t-4}, x_{1,t-3}, \ldots, x_{1,t},\ x_{2,t-4}, \ldots, x_{2,t},\ \ldots,\ x_{10,t-4}, \ldots, x_{10,t}\right]$$

$$\hat{y}_{s,t} = f_s(\mathbf{x}_t)$$

*Variant 2:*

In this variant, the lag values from only stop $s$ were used to train model $f_s$.

$$\hat{y}_{s,t} = f_s\left(x_{s,t-4},\ x_{s,t-3},\ x_{s,t-2},\ x_{s,t-1},\ x_{s,t}\right)$$

### 5.4 Combined Autoregressive Models

A single combined model $f$ was trained on data from every stop, at every time step. This model sees data from all stops and can exploit similarities in the data distribution between stops.

$$\hat{y}_{s,t} = f\left(x_{s,t-4},\ x_{s,t-3},\ x_{s,t-2},\ x_{s,t-1},\ x_{s,t}\right)$$

### 5.5 Integrated Models

Inspired by differencing in ARIMA models, integrated models were created by taking the four differences between each of the five lag values and using them as regressors (i.e. first order differencing):

$$d_{s,t} = x_{s,t-i} - x_{s,t-i-1}, \quad \text{for } i = 0, 1, 2, 3$$

As in the baseline model, the mean of the lags is calculated:

$$\mu_{s,t}^{(\text{lag})} = \frac{1}{5}\sum_{j=0}^{4} x_{s,t-j}$$

The difference between the mean of the lags and the value to be predicted is taken as the target for model $f$ to predict:

$$\Delta y_{s,t} = y_{s,t} - \mu_{s,t}^{(\text{lag})}$$

$$\hat{\Delta y}_{s,t} = f\left(d_{s,t},\ d_{s,t-1},\ d_{s,t-2},\ d_{s,t-3}\right)$$

The prediction is made by adding the predicted residual to the calculated mean of the lag values:

$$\hat{y}_{s,t} = \hat{\Delta y}_{s,t} + \mu_{s,t}^{(\text{lag})}$$

In cases of model instability (if $\left|\hat{\Delta y}_{s,t}\right|$ exceeds 100), $\hat{y}_{s,t}$ is predicted as $\mu_{s,t}^{(\text{lag})}$.

### 5.6 Integrated Models with Exogenous Variables

These models add a vector of exogenous variables $\mathbf{w}_t$ at each time step, and perform prediction as in the integrated model.

$$\hat{\Delta y}_{s,t} = f\left(d_{s,t},\ d_{s,t-1},\ d_{s,t-2},\ d_{s,t-3},\ \mathbf{w}_t\right)$$

In all cases, $\mathbf{w}_t$ includes a one-hot encoding of the station ID $\mathbf{s} \in \{0,1\}^S$. In the *seasonal integrated* model, $\mathbf{w}_t$ additionally has a time embedding:

$$\mathbf{w}_t = \begin{bmatrix} \sin\left(2\pi \cdot \frac{h_t}{24}\right) \\ \cos\left(2\pi \cdot \frac{h_t}{24}\right) \\ \mathbf{s} \end{bmatrix}$$

In the *time-aware* model, $\mathbf{w}_t$ consists of the time embedding as well as binarised features indicating whether it is the weekend, a public holiday, or whether an event was happening in Salvador.

In the *full exogenous variables* model, $\mathbf{w}_t$ consists of all features in the time-aware model, as well as every exogenous variable included in the dataset as above.

## 6 Results

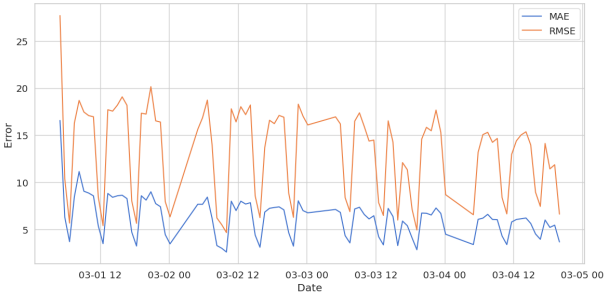
Table 2 shows the overall MAE and RMSE results for each model across all experiments.

The best performing model was a seasonal integrated model using a single FIMT-DD, which had an overall MAE of 6.24 and RMSE of 14.10 – the lowest error across all models tested. The plot for MAE and RMSE over time for this model is shown in Figure 1.

Table 2. MAE and RMSE for each experiment and model variant

| Experiment | Model | MAE | RMSE |
|---|---|---|---|
| *Baseline naive* | | 7.431 | 17.863 |
| Station + hour Time-aware baseline | KNN | 10.37 | 33.69 |
| | FIMT-DD | 20.85 | 49.11 |
| | ARF | 14.95 | 32.78 |
| Per-stop autoregressive model, variant 1 | KNN | 6.336 | 14.702 |
| | FIMT-DD | 12.316 | 33.171 |
| | ARF | 29.939 | 70.226 |
| Per-stop autoregressive model, variant 2 | KNN | 7.165 | 15.178 |
| | FIMT-DD | 9.583 | 22.152 |
| | ARF | 28.091 | 66.420 |
| Combined autoregressive model | KNN | 10.908 | 28.973 |
| | FIMT-DD | 8.287 | 21.071 |
| | ARF | 56.723 | 96.274 |
| Integrated model | KNN | 7.663 | 17.177 |
| | FIMT-DD | 6.855 | 14.892 |
| | ARF | 8.050 | 17.053 |
| Seasonal integrated model | KNN | 7.313 | 16.711 |
| | **FIMT-DD** | **6.240** | **14.104** |
| | ARF | 8.126 | 17.510 |
| Time-aware integrated model | KNN | 7.313 | 16.711 |
| | FIMT-DD | 7.422 | 15.532 |
| | ARF | 8.336 | 17.410 |
| Full exogenous variables integrated model | KNN | 7.347 | 16.642 |
| | FIMT-DD | 13.028 | 22.146 |
| | ARF | 8.165 | 18.883 |

Fig. 1. RMSE and MAE over time for best model

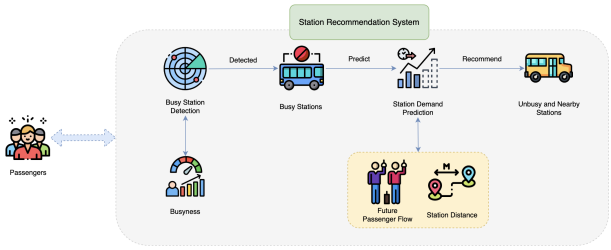



The best KNN model was the first variant of the per-stop autoregressive model, which had a MAE of 6.34 and a RMSE of 14.70. No ARF models were able to achieve lower error than the baseline naive model.

## 7 Applications

### 7.1 Station Recommendation System

Fig. 2. Station recommendation system architecture




We implemented a proof-of-concept station recommendation system (Figure 2) that demonstrates how predicted passenger load can be used in real time to assist passengers in avoiding crowded stops. This system uses our model to predict the average 5-minutely load for the next 30 minutes. If the stop is predicted to be crowded, our system attempts to find a nearby, less crowded stop that carries the same route. This may be useful for passengers who aim to avoid crowded stops and buses.

A simulation was run of the station recommendation system. It assumed all ten stops on the map carried the same routes, and that it would be feasible to walk between them. In practice, this would not necessarily be the case: this system is practically designed for stops that are close together. However, for the purposes of a proof-of-concept, we conducted the simulation on the ten stops in the provided dataset.

The system runs in parallel with any test-then-train streaming model. The model is used to predict the average load $\hat{y}_{s,t}$ at each stop over the next 30 minutes. Then, a predicted busyness index $\hat{b}_{s,t}$ is calculated, which takes into account both the absolute predicted load $\hat{y}_{s,t}$ and the normalised z-score of load for that stop $\hat{z}_{s,t}$ (using streaming estimates of mean $\hat{\mu}_{s,t}$ and standard deviation $\hat{\sigma}_{s,t}$).

$$\hat{z}_{s,t} = \frac{\hat{y}_{s,t} - \hat{\mu}_{s,t}}{\hat{\sigma}_{s,t} + 10^{-6}}$$

$$\hat{b}_{s,t} = \left(\frac{1}{1 + e^{-k_z(\hat{z}_{s,t} - z_{\text{thresh}})}}\right) \cdot \left(\frac{1}{1 + e^{-k_y(\hat{y}_{s,t} - y_{\text{thresh}})}}\right)$$

For the purposes of our simulation, we set our absolute threshold $y_{\text{thresh}}$ to 100, and our z-score threshold $z_{\text{thresh}}$ to 1.5, with slope constants $k_z$ of 3 and $k_y$ of 0.05. Any stop that had an absolute load greater than 100 and a z-score greater than 1.5 was considered busy. In practice, the calculation of busyness index could be amended to include variables such as the size of the station.

For busy stops at each time step, the five nearest stops were ranked depending on a suitability score $\hat{u}_{s_b,s_a,t}$ as calculated below. Ranking was dependent on the busyness index of the original busy stop $\hat{b}_{s_b,t}$, the busyness index of the alternate stop $\hat{b}_{s_a,t}$ (with load estimated using the streaming model), the Euclidean distance between them $d_{s_b,s_a}$, and a scaling factor $\alpha$. This factor represents the longest distance a passenger would be willing to walk to avoid a busy stop

(10 km in the case of our experiment, but likely lower in practice). In practice this metric should also take into account routes carried by the alternative stop and if they would get a passenger to their destination in more or less time.

$$\hat{u}_{s_b,s_a,t} = \hat{b}_{s_b,t} - \hat{b}_{s_a,t} - \frac{d_{s_b,s_a}}{\alpha}$$

If $\hat{u}_{s_b,s_a,t} < 0$ for all $s_a$, no recommendation was issued for stop $s_b$. Otherwise, the most suitable alternate stop (with highest $\hat{u}_{s_b,s_a,t}$) was recommended.

Example output from our proof of concept simulation, using the FIMT-DD seasonal integrated model, is shown in Table 3. An animation is provided in the supplementary material that demonstrates how the station recommendation system would work over the course of a weekday evening.

| Time | Busy Stop | Alternative (Distance) |
|---|---|---|
| 17:15 | Rua da Conceição da Praia, Comercio, Salvador | Largo Campo Grande, Salvador (1824m) |
| 17:20 | Av. Vale Do Tororo 327, Salvador | Largo Campo Grande, Salvador (1178m) |
| | Rua da Conceição da Praia, Comercio, Salvador | Largo Campo Grande, Salvador (1824m) |
| 17:25 | Estação Rodoviária | Av. Eng. Oscar Pontes, Calçada (4896m) |
| | Av. Vale Do Tororo 327, Salvador | Largo Campo Grande, Salvador (1178m) |
| | Rua da Conceição da Praia, Comercio, Salvador | Largo Campo Grande, Salvador (1824m) |

Table 3. Busy stops and nearest recommended alternatives.

### 7.2 Feature Importances

Mean-substitution feature perturbation methods were used to derive feature importances for the final exogenous variables model. For each prediction, each feature was set to its mean value, and the absolute change in the predicted load was observed. The importance values in Table 4 show the mean effect of this for each value across the dataset (omitting features with an importance of zero). The most important features were the one-hot encodings of the stop IDs, as well as the time of day encodings. Autoregressive features were also important. Weather features such as temperature and precipitation contributed somewhat, but were less important. There was minimal contribution from pollution features, and features that varied on a daily scale rather than hourly (such as Google Trend results, levels of flu or dengue, and day of the week).

Table 4. Mean absolute feature importances (non-zero)

| Feature | Mean Absolute Importance |
|---|---|
| Stop ID: 258781031 | 4.241931 |
| Stop ID: 43768720 | 1.057044 |
| Stop ID: 44783914 | 0.373962 |
| Time encoding: hour_sin | 0.144512 |
| Time encoding hour_cos | 0.039333 |
| Stop ID: 44783654 | 0.034803 |
| Stop ID: 230565994 | 0.023208 |
| Stop ID: 47568123 | 0.017710 |
| Stop ID: 125960550 | 0.015940 |
| Stop ID: 44072192 | 0.014185 |
| Stop ID: 44784438 | 0.013169 |
| Autoregressive: diff_0 | 0.013054 |
| Stop ID: 45833547 | 0.008432 |
| Autoregressive: diff_1 | 0.008153 |
| Ozone | 0.004516 |
| Temperature | 0.003651 |
| Precipitation | 0.003390 |
| Cloud Cover | 0.001313 |
| Wind Speed | 0.000267 |
| Autoregressive: diff_3 | 0.000085 |

## 8 Discussion

In this study, we adapted principles of ARIMA to the streaming setting. Autoregression was key to our approach: almost all autoregressive approaches outperformed the baseline that did not use autoregressive features. Of the autoregressive models, the best model was the FIMT-DD seasonal integrated model. Compared to a simple autoregressive model, training on the differences and predicting the residual was able to produce better predictions. By predicting deviations from the recent mean, this method is more robust to non-stationarity.

Adding an embedding of the time and an embedding of the station ID were able to reduce the error, but further exogenous variables were not able to improve the accuracy. We did not add the moving average component of ARIMA to our model as this component becomes unstable in the streaming setting [Kozitsin et al. 2021].

FIMT-DD is a decision tree model created for streaming data, and it is the regression analogue of the Hoeffding tree. It offers fast and accurate performance, whilst being robust to concept drift through its drift detection and adaptation mechanisms [Ikonomovska et al. 2011]. The best performance with the seasonal integrated model was achieved with FIMT-DD.

The delayed labelling setting increased the difficulty of this problem. Autoregression often relies heavily on the labelling of the most recent time points in order to achieve high accuracy, however this was not available in this case. The models had to effectively extrapolate up six time points ahead, which made it difficult to achieve low error. Nevertheless, we were able to create a final model which achieved substantially better performance than the baseline naive model.

The station recommendation system provides an example application of our model. It has demonstrated that the predictions issued by our model are able to create sensible predictions that are responsive to changes in the data over time. In practice, additional factors such as bus routes, walking routes, transfer times, and the capacity of each station would need to be taken into account. The current proof-of-concept assumes all stations carry the same routes and are walkable: these assumptions could be addressed in future work with access to richer transport network data.

## 9 Conclusion

This study has optimised and tested autoregressive streaming algorithms for prediction of bus load at ten bus stops in Salvador, Brazil. The best model was a seasonal integrated model using FIMT-DD, inspired by seasonal ARIMA models. We demonstrated that this model has potential for practical use, via a station recommendation system.

## 10 CRediT Statement

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